Content

Exam 1 will cover topics from Lecture 1-15. The specific list of topics is the following:

Lectures 1-7

1. Ordered Fields
    - Examples/Non-examples
    - Definition and properties of $\leq$
2. The Real Line
    - Definition and properties of the absolute value
    - Definition and properties of sup/inf
    - The completeness axiom
    - Archimedean property
    - Dense sets
3. Metric Spaces
    - Examples of metric spaces.
    - Norms induce metrics
    - Open/Closed sets in metric spaces
    - Bounded sets.
    - Interior, Closure, and Boundaries
    - Equivalent norms and metrics
4. Sequences in Metric Spaces
    - Bounded and convergent subsequences
    - Subsequences
5. Sequences in $\mathbb{R}$
    - Properties of convergent subsequences
    - Monotone sequences
    - Definition and properties of limsup/liminf
6. Completed Metric Spaces
    - Cauchy sequences
    - Definition of a complete metric space
    - $\mathbb{R}$ is complete
    - $\mathbb{R}^d$ is complete
7. Topological Spaces
    - Definition of a Topological Space $(X, \mathcal{T})$.
    - Metric Topology
    - Convergence on Topological spaces
    - Definition and properties of Compact sets

Practice Questions of Lectures 1-7

1. For each question, say whether such an object exists. If it does, give an example. If it does not, give a short explanation on why not. Assume all the sequences are in $\mathbb{R}$.
   (a) A sequence which is not monotone and has no convergent subsequence.
   (b) A Cauchy sequence with a divergent subsequence.
   (c) A sequence with exactly two subsequential limits.
   (d) A bounded sequence with no Cauchy subsequence
2. True or False.
   (a) If (x_n) and (y_n) diverge, then the sequence $x_n y_n$ diverges.
   (b) If $(x_n) \to 0$ and (y_n) is bounded then $x_n y_n \to 0$.
   (c) Arbitrarily unions of closed sets are closed.
   (d) Closed and bounded sets are compact in metric spaces.
   (e) Let S be a nonempty bounded subset of $\mathbb{R}$ then $\inf S < \sup S$.
3. Prove the lower triangle inequality using the triangle inequality.
4. Use the fact that $\mathbb{Q}$ is dense to prove that there are infinitely many rationals in $[a, b]$ for $a, b \in R$.
5. Assuming that $\mathbb{Q}$ is dense show that for all $x \in \mathbb{R}$ there exists an n such that $n \leq x \leq n+1$.
6. Let X be a set of real numbers that has a supremum, and let $a := \sup X$. Prove that $X \cap (a - \varepsilon, a] \neq \emptyset$ for every $\varepsilon > 0$.
7. Prove that
$$\limsup_{n\to\infty}(s_n + t_n) \leq \limsup_{n\to\infty} s_n + \limsup_{n\to\infty} y_n$$
8. Suppose that a_n and b_n are Cauchy sequences. Prove that $(a_n b_n)$ is a Cauchy sequence without using the fact that Cauchy sequences converge.
9. Let A and B be subsets of a metric space E. Prove that $\overline{A \cup B} = \overline{A} \cup \overline{B}$
10. Let $f : \mathbb{R} \to \mathbb{R}$ such that
$$|f(x) - f(y)| \leq |x - y|^2.$$
Prove that f is continuous, but not uniformly continuous.
11. Prove that for a bounded real sequence s_n that if $s_{n_k} \to L$ as $k \to \infty$ then
$$\liminf s_n \leq L \leq \limsup s_n.$$
12. Prove that for a real sequence s_n, $s_n \to \infty \iff s_n^{-1} \to 0$.
13. Let s_n be a sequence in a Metric space (E, d). Prove that if $\forall n$
$$d(s_n, s_{n+1}) \leq d(s_n, s_{n-1})$$
is Cauchy.
14. Prove that $\lim n^{1/n} \to 1$.
15. Find the limit of
$$s_n = \frac{n^3 + 6n^2 + 7}{4n^3 + 3 * n - 4}$$
without using theorems from Calculus.